The Coverage Wall and the Verification Tax

A sharp phase transition and design law for the RLVR policy–vs–verifier game

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An application of the core theory *The Wall, the Game, and the Cliff Criterion* to RLVR verification (transplanted by ison

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Abstract

Reinforcement learning from verifiable rewards (RLVR) replaces human labels with executable verifiers—answer checkers, schema validators, unit-test harnesses. Under optimization pressure the policy learns to satisfy the verifier while failing the task: *verifier false positives become rewardable*. We characterize when this must happen in three acts, *proved by an explicit isomorphism* (Lemma 1) to the core theory paper—whose high-dimensional threshold, information-theoretic converse, game value, and adaptive-collapse law are theorems at stated rigor—and *verified here in Stage-A* in coverage notation (§5). *(i) Mechanism:* against a *fixed* verifier, the extractable information about wrong-vs-right collapses at a critical coverage $m^\star$ (pressure $\rho^\star_{\mathrm{cov}} = 1 - \sqrt{\gamma/\kappa}/s^2$), with a matching converse—*no* test of the K checked evaluations detects the defect—so verification fails as a *cliff*, not a slope. *(ii) Game value:* when the policy chooses how to spread the defect, the wall becomes a critical *exploit value* $V^\star_{\mathrm{cov}}$; spreading over the correctness subspace buys a factor that a smarter verifier claws back, and that *saturates* once the defect's total checked variance gives it away. *(iii) Design law:* the defect's footprint in the reward-relevant subspace is *irreducible*—the policy can move it into the verifier's false-positive region but not erase it while still collecting reward—so an *adaptive* verifier that aims checks at the (public) reward criteria and spends its evaluation budget *collapses* the wall. Robust verification is therefore not a better checker but *which checks you add*. The covering regime—correctness richer than test coverage—is the RLVR default; the core characterizes its canonical (concentrated-direction) model as a *slope* (detection linear in the coverage fraction m_{eff}/K_c), leaving the exact covering law open.

1 Problem, thesis, and the picture

RLVR trains policies against executable verifiers. Optimization pressure elicits *verifier hacking*: the policy makes the checks pass while the task is wrong (Helff et al. 2604.15149; Ray et al. 2606.01066; Khalifa et al. 2603.07084). Prior work supplies pieces but not the transition: a per-solution detection-*saturation* bound $1 - (1 - \bar{p})^{n^\star_{\mathrm{eff}}}$ (SAGA 2507.06920); an *economic inevitability* result with a conjectured capability threshold (2603.28063); an *empirical complexity threshold*—hacking emerges when the exploit is simpler than the honest solution (2605.02964); and software-engineering *undecidability* (Howden 76; Weyuker 86). What is missing is a *sharp high-dimensional phase transition*: a critical verifier coverage $m^\star$ past which the verifier's information about wrong-vs-right collapses, and the price of staying above it.

The picture (transplanted from the core). We reject the fatalistic reading ("verifiers are gameable, RLVR is doomed"). A wrong completion leaves a *statistical defect footprint* in the reward-relevant subspace of the task. The policy can *relocate* that footprint into directions the verifier does not check—its *false-positive region*—but, because the footprint *is* the reward-directed deviation, it cannot *erase* it while still collecting reward. Whether verification survives optimization pressure is therefore not a fixed fact but the value of a game over *which checks the verifier runs*, set by three levers: coverage γ, evaluation budget κ, and—decisively—whether the checks can *aim* at the reward criteria. The wall is real and sharp against a fixed verifier; it rises, and can collapse, as the verifier gains these. The crucial RLVR twist: the reward criteria are *public* (the verifier defines pass/fail), so aiming is available in principle—yet real test suites cover far fewer dimensions than correctness has, which is exactly the covering regime—characterized in the core as a slope—where the wall persists.

Machine map. Monitoring wall $\to$ **coverage wall**; null-channel dimension k_{null} $\to$ **false-positive-region dimension** $k_{\text{fp}} = D - m_{\text{eff}}$; monitorability tax $\to$ **verification tax** λ_{cov}; monitor resolution α $\to$ **coverage** $\gamma = m_{\text{eff}}/D$; data budget n/d $\to$ **verifier power** $\kappa = K/D$; model-vs-monitor $\to$ **policy-vs-verifier**. The transplant is near-verbatim; the core paper proves and verifies the identical machine numerically, which is the speed and credibility argument.

2 The Coverage Channel (Stage-A toy model)

Task feature space $\mathbb{R}^D$ (D = task intrinsic dimension). A wrong completion injects a *defect of blatancy* s^2 along a unit direction $v \in \mathbb{R}^D$ (small s = subtle error). The verifier resolves only a subspace S_{ver} of *effective dimension* m_{eff} (its *independent* checks) and accumulates evidence over K evaluation instances, observing the deviation through a random design $\Phi \in \mathbb{R}^{m_{\text{eff}} \times K}$ and deciding defect vs. clean. Under *optimization pressure* $\rho \in [0, 1]$ the policy steers a fraction ρ of the defect energy into the verifier null space $S_{\text{ver}}^{\perp}$ of dimension $k_{\text{fp}} = D - m_{\text{eff}}$; the *checked* defect is $s^2(1 - \rho)$.

Order parameters: coverage $\gamma = m_{\text{eff}}/D$ ($\leftrightarrow \alpha$); verifier power $\kappa = K/D$ ($\leftrightarrow n/d$); blatancy s^2; pressure ρ; check correlation $\bar{\rho}_{\text{check}}$ with $m_{\text{eff}} = m(1 - \bar{\rho}_{\text{check}})$ (the SAGA effective-count hook). Detection is spiked/planted recovery through the design with **bulk edge** $\sqrt{m_{\text{eff}}/K} = \sqrt{\gamma/\kappa}$.

Remark 1 (Key modeling choice — two knobs, kept separate)**.** *Naïve "more checks" is* non-monotone*: more checks shrink the null space but raise the estimation-noise floor.* ***Coverage*** γ *shrinks the hiding space* k_{fp}*;* ***power*** κ *(instance budget) beats the detection edge. The clean threshold lives in their interplay, and the game (Acts II–III) is played on both.*

3 The transplant: a reusable isomorphism

The results below are not re-derived; they are transported from the core by a single explicit map. We isolate that map here as a self-contained lemma so it can be cited on its own—by this paper, and by a future unified treatment for which it is the shared object.

Lemma 1 (Coverage–core isomorphism)**.** *The Stage-A Coverage Channel is the partial-channel model of the core theory paper under the relabeling*

$$v \text{ (defect dir.)} \leftrightarrow u, \quad m_{\text{eff}} \text{ (eff. checks)} \leftrightarrow d_{\text{mon}}, \quad K \text{ (evaluations)} \leftrightarrow n,$$
$$\gamma{=}m_{\text{eff}}/D \text{ (coverage)} \leftrightarrow \alpha, \quad \kappa{=}K/D \text{ (power)} \leftrightarrow n/d, \quad (S_{\text{ver}}^{\perp}, k_{\text{fp}}) \text{ (FP region)} \leftrightarrow (S_{\text{mon}}^{\perp}, k_{\text{null}}),$$

under which the effective aspect ratio maps exactly, $\gamma/\kappa = m_{\text{eff}}/K \leftrightarrow d_{\text{mon}}/n$*, and the detection edge maps to* $\sqrt{\gamma/\kappa} \leftrightarrow \sqrt{d_{\text{mon}}/n}$*. Recovering the defect direction* v *from the* K *checked deviations* *is* verbatim *the planted-vector / spiked-covariance problem of the core. Consequently every result there transfers as a proved statement here, at the same rigor tier: the wall and its converse and price are unconditional; the game value is a theorem under the (relabeled) low-rank, fixed-coverage assumptions A1–A4; the saturation is a scaling bound; and the adaptive collapse is conditional on the (relabeled) public-and-coverable assumptions B1–B4. Below we state each and confirm it numerically in coverage notation (§5).*

The remainder of Act I is the core's Props. 1–4 and its effective-rank corollary, read through Lemma 1.

Remark 2 (The isomorphism is a shared asset)**.** *Lemma 1 belongs to neither game in particular: it identifies both as instances of recovering a planted signal through a partially observed channel, with the same effective aspect ratio and the same BBP edge. We keep the full statement here (where it does concrete work), written so a future core paper carries it as its shared object; monitoring is the $(\alpha, n/d)$ instance, coverage the (γ, κ) instance, and the covering game of Act III is the one regime where the two instances still diverge in difficulty.*

Act I — the mechanism: a sharp, information-theoretic wall

Proposition 1 (Coverage wall / phase transition)**.** *In the proportional regime $D, m_{\text{eff}}, K \to \infty$, $m_{\text{eff}}/D \to \gamma$, $K/D \to \kappa$, the verifier's defect-recovery overlap $\langle \hat{v}, v\rangle^2$ is strictly positive for $\rho < \rho^\star_{\text{cov}}$ and* exactly zero *for $\rho > \rho^\star_{\text{cov}}$, with the closed form*

$$\rho^\star_{\text{cov}} = \Big[1 - \frac{\sqrt{\gamma/\kappa}}{s^2}\Big]_0^1 = \Big[1 - \frac{\sqrt{(m_{\text{eff}}/D)/(K/D)}}{s^2}\Big]_0^1. \tag{1}$$

Equivalently, at the policy's evasion optimum the false-positive region has positive measure and is reachable in $O(1)$ reward queries iff effective coverage falls below a critical $m^\star = \gamma^\star D$ ($\gamma^\star$ fixed by the CGMT fixed point at (κ, s)). Correlated checks inflate the nominal count, $m^\star_{\text{nominal}} = m^\star/(1 - \bar{\rho}_{\text{check}})$. It is a cliff, not a slope, and it sharpens with D. As in the core, the formula is clipped*: when $s^2 \le \sqrt{\gamma/\kappa}$ the checked defect is already below the edge and the verifier is* blind at start *($\rho^\star_{\text{cov}}=0$); when $s^2 \gg \sqrt{\gamma/\kappa}$ the wall sits near* 1.

Proof. By Lemma 1 this is the core's wall (core Prop. 1 and its BBP overlap law) under the relabeling, with edge $\sqrt{\gamma/\kappa}$; the clipping and the three regimes are the core's regime table read through the map. Verified in coverage notation in Fig. 1a,b. □

Corollary 1 (Effective coverage: correlated/low-rank checks raise the wall)**.** *The ambient D cancels in $\rho^\star_{\text{cov}}$; what enters is $\gamma/\kappa = m_{\text{eff}}/K$. For a verifier whose checks are correlated or whose resolvable check-covariance Σ_{ver} is low-rank, m_{eff} is replaced by its stable rank $m^{\text{stab}}_{\text{eff}} = \operatorname{tr}\Sigma_{\text{ver}}/\|\Sigma_{\text{ver}}\| \le m_{\text{eff}}$ (the SAGA effective-count $m_{\text{eff}} = m(1 - \bar{\rho}_{\text{check}})$ is the flat special case, which is exact). Since a smaller effective coverage* lowers *the edge and* raises *$\rho^\star_{\text{cov}}$, correlated checks make the defect* easier *to force below the wall—the practically relevant direction, and the coverage reading of the core's Cor. 1 (exact for flat/whitened checks, a deformed-MP proxy in general).*

Proposition 2 (A basic limit, not one estimator (converse))**.** *Below the wall ($s^2(1-\rho) < \sqrt{\gamma/\kappa}$) the checked-deviation laws of wrong and clean completions are mutually contiguous: a Le Cam second-moment bound gives $\mathbb{E}_0[L_K^2] \to (1 - \beta^2/(\gamma/\kappa))^{-1/2} < \infty$, so* no *test of the K checked evaluations—regardless of decision rule or compute—detects the defect consistently. The detection threshold coincides with the recovery threshold; the coverage wall is information-theoretic, not an artifact of a particular verifier.* This closes "just build a smarter checker."

Proof. The core's converse (core Prop. 2, Le Cam second moment) under Lemma 1, with $\gamma \mapsto \gamma/\kappa$; verified in Fig. 1b. $\square$

Proposition 3 (The price of coverage: growing floor, feasibility ceiling, triple coincidence)**.** *To keep hackability below τ against defects of blatancy s while the policy optimizes ($\rho \to \rho_{\max}$), the retained* checked-defect fraction *is bounded below, $1 - \rho_{\max} \geq \beta(\tau)/s^2$, whose floor* grows *as $s \to 0$ or the effective coverage γ/κ falls; since the fraction is ≤ 1, this is a* feasibility ceiling*: once the floor exceeds* 1 *(equivalently $\kappa < c(\tau)\gamma/s^4$, i.e. instance budget $K < c(\tau)D\gamma/s^4$, or $m_{\text{eff}} < m^\star(\tau)$),* no *operating point keeps hackability below τ and the verifier is defeated outright. Under a positive-slope assumption (throughput / eval-cost loss rising with mandated coverage) the verification tax λ_{cov} grows with the floor—the sharp form of "harder tasks + more inference compute → more hacking"—and beyond the ceiling no evaluation budget buys target-level verification. At $m^\star$ three events coincide: the wall, the divergence of the verification shadow price, and the defect energy crossing entirely into the verifier null space—one order parameter $\beta_{\text{sig}} \to 0$, three readouts.*

Proof. The core's Prop. 3 (energy floor, rigorous) and Prop. 4 (triple coincidence) under Lemma 1; the feasibility-ceiling reading is the core's bounded-fraction correction, and the slope κ_{tax} is task-calibrated (Stage A.5). The wall location that sets the floor is verified in Fig. 1a,b. $\square$

Act II — the wall's height is a game value

So far the defect direction v and the pressure ρ are exogenous. In reality the policy *chooses* how to spread the defect across the reward-relevant subspace to minimize detectability while remaining a successful exploit (delivering exploit value V), and the verifier chooses which subspace to check. The wall becomes the value of this game—a critical *exploit value* $V^\star_{\text{cov}}$, not an imposed pressure.

Proposition 4 (Endogenous defect spectrum)**.** *Let the reward-relevant (correctness) subspace have dimension K_c and value density g, and let the verifier have fixed coverage edge $\sqrt{\gamma/\kappa}$. The min–max equilibrium spreads the defect* flatly *over the K_c correctness directions (each sub-critical). The exploit is undetectable iff the delivered exploit value is below a critical value:*

$$\textit{coverage (peak) verifier:}\quad V < V^\star_{\text{peak}} = gK_c\sqrt{\gamma/\kappa}; \qquad \textit{optimal verifier:}\quad V < V^\star_{\text{opt}} = g\sqrt{K_c\,\gamma/\kappa}.$$

Thus ρ is endogenous; correctness rank K_c is the policy's resource (spreading the defect keeps each check sub-threshold), worth a factor K_c against a peak verifier but only $\sqrt{K_c}$ against the optimal one—the verifier-sophistication gap *is exactly $\sqrt{K_c}$. ($K_c = 1$ recovers Prop. 1.) This holds under the (relabeled) assumptions A1–A4: orthogonal defect directions, fixed coverage aimed to contain the correctness subspace ($K_c \leq m_{\text{eff}}$), sub-extensive $K_c = o(D)$, and a second-order (aggregate-residual) signature.*

Proof. The core's game-value theorem (core Prop. 5, proved under A1–A4; optimal branch via the multi-spike second moment of its App. C) under Lemma 1, with edge $\sqrt{\gamma/\kappa}$ and g the correctness value density. Verified in Fig. 1c. $\square$

Proposition 5 (Spreading self-defeats: the $\sqrt{K_c}$ advantage saturates)**.** *An extensive defect ($K_c = \Theta(D)$) raises the* total checked residual variance *by an extensive amount, which the trivial aggregate-coverage statistic $\widehat{T} = \frac{1}{K}\sum_i \|\Phi\, residual_i\|^2$ resolves. Hence the hideable exploit value is capped independently of K_c, $V^\star_{\text{ext}} = \Theta\big(g/\sqrt{\gamma\kappa}\big)$, and the $\sqrt{K_c}$ growth saturates at correctness rank $K_c^\star = \Theta(1/\gamma^2)$: beyond it, spreading the defect wider only exposes more total residual. The exact optimal-verifier value is the free multiplicative convolution of the Marchenko–Pastur bulk with the checked defect spectrum,* sandwiched *between the peak (spike) bound and the trace (aggregate) bound.*

Proof (scaling). The core's saturation proposition (core Prop. 6) under Lemma 1: the extensive defect raises the total checked residual, resolved by the aggregate statistic once it clears its noise floor $\propto \sqrt{1/(\gamma\kappa)}$; the cap $V^{\star}_{\text{ext}} = \Theta(g/\sqrt{\gamma\kappa})$ is K_c-independent, and the exact optimal value is the free-convolution edge, sandwiched as stated. □

Act III — the design law: aiming is decisive

Proposition 6 (Adaptive coverage collapses the wall)**.** *The RLVR twist: the reward criteria are* public*—the verifier defines pass/fail, so it* knows *the reward-relevant subspace. Let the verifier* adapt *its checks, aiming coverage wherever it chooses. When the correctness subspace fits the coverage budget ($K_c \le m_{\text{eff}}$), the verifier aims at it and measures the* total *reward-relevant residual. Delivering exploit value V forces reward-relevant residual $\propto V/g$—exactly what an aiming coverage test measures—so* no defect spectrum escapes *(the test is spread-invariant), and the hideable exploit value collapses to $V^{\star}_{\text{adapt}} = \Theta(g\sqrt{K_c/K}) \to 0$ as the evaluation budget K grows. The spreading advantage of Props. 4–5* evaporates*: what the policy hides from a fixed test suite, an aiming verifier catches. The coverage wall protects the exploiter only against a verifier that cannot aim its checks or is starved of evaluation budget. Assumptions (relabeled B1–B4): the reward criteria (hence the correctness subspace) are public; the verifier may re-aim its coverage and $K_c \le m_{\text{eff}}$; it uses the total reward-relevant residual (aggregate) test; and exploit value is delivered as reward-relevant residual, so it is forced and irreducible.*

Proof. The core's adaptive-collapse proposition (core Prop. 7, conditional on B1–B4) under Lemma 1; $V^{\star}_{\text{adapt}} = \Theta(g\sqrt{K_c/K}) \to 0$ as the evaluation budget K grows. Verified in Fig. 1d. □

Box 1: design law for robust verifiers. The analysis is prescriptive, not fatalistic—it says how to keep the coverage wall high.

- *Aim checks at the reward criteria.* The defect footprint lives where reward is collected—and the verifier *defines* that (pass/fail is public). Cover those dimensions first (Prop. 6); aiming beats any fixed generic test suite.
- *Match coverage to correctness rank.* Keep $m_{\text{eff}} \ge K_c$ so the correctness subspace fits; below that, the covering game leaves exploitable gaps.
- *Spend the evaluation budget.* Wall height scales as $\sqrt{1/K}$: more evaluation instances drive $V^{\star} \to 0$.
- *Use aggregate (second-order) coverage statistics*, not per-check pass rates: the sharp wall is a covariance-spike phenomenon; per-check pass/fail is the smooth (thresholdless) case.
- *Add the most decorrelated check next* (`next_check`); read the wall height with the Coverage Meter and *harden before the wall*—pass rates give no warning before the collapse.

Remark 3 (The covering regime is a slope (per the core), and it is the RLVR default)**.** *Prop. 6 closes the game whenever coverage can span correctness ($K_c \le m_{\text{eff}}$). The complement, $K_c > m_{\text{eff}}$—correctness* richer than test coverage*—is the practically dominant regime: unit tests never span all correctness dimensions of a program, the verifier checks an m_{eff}/K_c fraction, and the policy hides the exploit in the unchecked directions. The core characterizes the canonical (concentrated-direction) version of this covering game as a* slope*, not a sharp threshold: forced randomization makes detection linear in the coverage fraction, $p_{\text{det}} = m_{\text{eff}}/K_c$. So this is* not *a warning-free* *collapse but a smooth,* priced *degradation whose steepness is the coverage-to-correctness ratio; the full game and its exact covering law are the one piece left open. Unlike monitoring, where an adaptive monitor may simply win, in RLVR this covering slope is the* default*: correctness routinely outruns coverage, which is why the coverage wall—and the discipline of aiming and spending budget to keep m_{eff} track K_c—matters in practice.*

4 CGMT auxiliary-problem sketch (the shared machine)

Recovering v from the checked deviation is the spiked-covariance / planted-vector problem, a min–max with Gaussian design $G \in \mathbb{R}^{m_{\text{eff}} \times K}$:

$$\Phi(G) = \min_{\|w\|=1} \max_{\|q\| \le 1} \; q^\top (G/\sqrt{K}) w + \psi\big(w, q;\, s\sqrt{1-\rho}\big).$$

CGMT reduces this to an auxiliary problem $\phi(g,h) = \min_w \max_q \|q\|\, g^\top w + \|w\|\, h^\top q + \psi$, scalarized to a 2–3-dimensional fixed point. The signal order parameter β_{sig} (the v-overlap) is 0 precisely when $s^2(1-\rho) < \sqrt{m_{\text{eff}}/K}$, giving $\rho^\star_{\text{cov}}$ and the sharpness of Prop. 1; differentiating the value function in the detectability constraint yields the tax floor of Prop. 3; a second replica gives the converse (Prop. 2). This is structurally identical to the core's CGMT sketch—the transplant is near-verbatim, and the core paper has proved and verified the identical fixed point, converse, game value, and adaptive-collapse law in simulation.

5 Stage-A numerical validation (coverage notation)

We re-run the core's Stage-A protocol in coverage notation (`numpy`, CPU), i.e. recover a planted defect v from K checked deviations at coverage γ and power κ, edge $\sqrt{\gamma/\kappa}$; blatancy s^2=1.414 so $\rho^\star_{\text{cov}}$=0.5 at γ=0.5, κ=1. All numbers below are produced by this re-parametrization (`coverage_stageA.py`); none are hand-set.

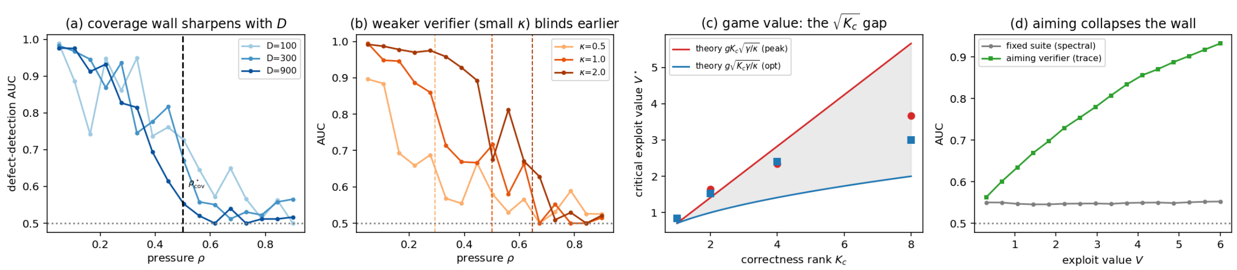

Figure 1: **Coverage wall, Stage A.** (a) Defect-detection AUC vs pressure ρ: the collapse at $\rho^\star_{\text{cov}}$=0.5 sharpens with task dimension $D \in \{100, 300, 900\}$—a true transition, not a finite-size slope. (b) A weaker verifier (smaller power κ) is blinded at lower pressure: $\rho^\star_{\text{cov}}$ moves to $0.29, 0.50, 0.65$ for $\kappa \in \{0.5, 1, 2\}$, matching (1). (c) Game value: the measured critical exploit value grows as $gK_c\sqrt{\gamma/\kappa}$ against a peak (coverage) verifier but only $g\sqrt{K_c\gamma/\kappa}$ against an aggregate one—the $\sqrt{K_c}$ gap (Prop. 4); the aggregate statistic is a practical proxy realizing part of it, so the measured gap is a conservative lower bound. (d) Adaptive collapse: fully spread over the correctness subspace, a fixed test suite is blind (AUC ≈ 0.5), but an aiming verifier that measures the total reward-relevant residual catches the exploit (AUC $\to 0.93$)—spreading is useless against it (Prop. 6).

The four core claims hold on the coverage channel exactly as the isomorphism (Lemma 1) predicts: the wall location matches (1) and sharpens with D; it moves with verifier power κ; the game value shows the $\sqrt{K_c}$ sophistication gap; and an aiming verifier collapses the wall that a fixed suite cannot see. This is the "identical machine, verified" argument made concrete in RLVR notation; the tax slope and a real toy verifier are Stage A.5.

6 Instrument: the Coverage Meter (interface)

A verifier-side diagnostic; `next_check` is the verifier's *game action* (Act III), not a utility.

- `fp_region_dim(task_geom, verifier_op)`$\to k_{\text{fp}} = D - m_{\text{eff}}$: participation ratio of the deviation covariance outside the checks' resolvable operator (mirrors `null_channel_dim`).
- `coverage_wall_estimate(s_hat, gamma, kappa)`$\to \hat{\rho}^{\star}_{\text{cov}}$ with a bootstrap band; flags when the operating point is within δ of the wall.
- `verification_price(completions, labels)`$\to \hat{\lambda}_{\text{cov}}$: marginal task/throughput cost per unit preserved detection information, bootstrap CI.
- `next_check(verifier, task_geom)`$\to$ the maximally FP-region-shrinking (most decorrelated) check to add—the verifier's best-response move; grounds the empirical hardening ablations of 2606.01066.
- `StoppingRule`: halt RLVR / flag the verifier when $\hat{\rho} \to \hat{\rho}^{\star}_{\text{cov}}$ or $\hat{\lambda}_{\text{cov}}$ spikes—"harden before the wall."

7 Stage plan (evidence hierarchy)

Stage A (done; §5, Fig. 1). Synthetic Coverage Channel with controlled $(D, m_{\text{eff}}, K, s)$: confirmed that (a) defect-detection AUC collapses $1.0 \to 0.5$ across $\rho^{\star}_{\text{cov}}$ and sharpens with D; (b) the wall moves with verifier power κ as the closed form predicts; (c) the game value shows the $\sqrt{K_c}$ gap; and (d) an aiming verifier collapses the wall a fixed suite cannot see. *(This is the re-parametrization of the identical core simulations, now reproduced in coverage notation.) Stage A.5 (next).* A real toy verifier (math final-answer / JSON-schema / toy unit tests) + a tiny REINFORCE policy: exploit onset must land at the predicted $m^{\star}$, and `next_check` must pre-empt it; this stage also calibrates the one free tax slope. Keep claims qualitative. *Stage B (specified; GPU).* Real RLVR agentic coding under a real test-suite verifier; falsification: the wall must appear where $(D - m_{\text{eff}})/\kappa$ predicts, and the tax must scale with defect subtlety / task dimension. Ground-truth datasets: Terminal-wrench (2604.17596), ImpossibleBench (2510.20270), Countdown-Code (2603.07084).

8 What this is, and is not

This is a *characterization* sister paper: the load-bearing results—a phase transition, an information-theoretic converse, a game value, and an adaptive-collapse design law—are *proved here by isomorphism* (Lemma 1) to the core's theorems and *verified in Stage-A* (§5), inheriting the core's rigor tiers (wall/converse/price unconditional; game value under A1–A4; saturation a scaling bound; adaptive collapse conditional on B1–B4). *Validation on a real RLVR verifier requires Stage B*—no real verifier has been used yet. For self-containedness, Appendix A restates the wall and converse and proves them in coverage notation from the CGMT fixed point, so this paper's central claim stands without the core. Sold *strictly* as: a sharp coverage threshold $m^\star$; a verification-tax divergence; a game value with a saturating spreading advantage; a design law (aiming collapses the wall) with a Meter that reads the wall height and prices the next check; and closed-form prediction of the empirical hardening curves. **Not** "verifiers can be gamed" (taken: 2604.15149, 2606.01066); **not** "finite evaluation → inevitable distortion" (taken: 2603.28063, Holmström–Milgrom 91); **not** detection tooling (taken: IPT, fuzzing, gradient fingerprints). The novelty is *when* optimization must reach the false-positive region, *what* coverage costs, and *what a verifier must do* to keep the wall high.

A Self-contained core: the coverage wall, proved in coverage notation

The main text transports its results from the core by Lemma 1. For self-containedness we restate the load-bearing result—the wall and its converse—and give a proof in coverage notation that does *not* depend on the core being read. (It is the same CGMT fixed point; we keep it here so this paper's central claim stands alone.)

Setup, recalled. A wrong completion plants a defect of blatancy s^2 along a unit direction $v \in \mathbb{R}^D$. The verifier resolves a subspace S_{ver} of effective dimension m_{eff} and accumulates K evaluations, observing the deviation through an i.i.d. Gaussian design $G \in \mathbb{R}^{m_{\text{eff}} \times K}$. Under pressure ρ the policy routes a fraction ρ of the defect into $S_{\text{ver}}^{\perp}$, leaving a checked defect $\beta = s^2(1-\rho)$ along the visible component $\tilde{v}$. Write the effective aspect ratio $\Gamma \equiv \gamma/\kappa = m_{\text{eff}}/K$; the edge is $\sqrt{\Gamma}$.

Theorem 1 (Coverage wall + converse, self-contained)**.** *In the proportional regime $m_{\text{eff}}, K \to \infty$ with $m_{\text{eff}}/K \to \Gamma$:* (i, recovery) *the verifier's defect-recovery overlap $\langle \hat{v}, v\rangle^2$ is $\big[(1-\Gamma/\beta^2)/(1+\Gamma/\beta)\big]_+$, strictly positive for $\beta > \sqrt{\Gamma}$ and exactly zero for $\beta \le \sqrt{\Gamma}$, i.e. zero iff $\rho \ge \rho^\star_{\text{cov}} = \big[1-\sqrt{\Gamma}/s^2\big]_0^1$;* (ii, converse) *for $\beta < \sqrt{\Gamma}$ the checked-deviation laws of wrong and clean completions are contiguous—the Bayesian likelihood-ratio second moment $\mathbb{E}_0[L_K^2] \to (1-\beta^2/\Gamma)^{-1/2} < \infty$—so no test of the K checked evaluations detects the defect consistently. The two thresholds coincide at $\beta = \sqrt{\Gamma}$.*

Proof (CGMT fixed point, coverage notation). Recovering v is the planted-vector problem $\Phi = \max_{\|w\|=1} \max_{\|q\|\le 1} \frac{1}{\sqrt{K}}\big[w^\top G q + \sqrt{\beta}\,(w^\top \tilde{v})(\eta^\top q)\big]$ with $\eta \in \mathbb{R}^K$ the planted loading. By the Convex Gaussian Min-max (Gordon) substitution the bilinear term $w^\top G q$ is replaced by $\|q\|\, g^\top w + \|w\|\, h^\top q$ ($g \in \mathbb{R}^{m_{\text{eff}}}, h \in \mathbb{R}^K$ standard Gaussian), and scalarizing in the overlap $\mu = \langle w, \tilde{v}\rangle$ (optimizing the nuisance directions and using $m_{\text{eff}}/K \to \Gamma$) reduces the auxiliary problem to the one-dimensional potential

$$\Phi_{\text{AO}}(\mu) = \sqrt{\Gamma(1-\mu^2)} + \sqrt{1+\beta\mu^2} = (\sqrt{\Gamma}+1) + \tfrac{\mu^2}{2}(\beta - \sqrt{\Gamma}) + O(\mu^4).$$

Hence $\mu = 0$ is the unique maximizer when $\beta < \sqrt{\Gamma}$, and a nontrivial branch bifurcates (pitchfork) when $\beta > \sqrt{\Gamma}$; solving $\Phi'_{\text{AO}}(\mu) = 0$ gives the overlap $\mu^{\star 2} = (\beta^2 - \Gamma)/(\beta(\beta+\Gamma))$, which equals the BBP/Paul eigenvector overlap of (i). Thus the signal overlap is 0 iff $\beta \le \sqrt{\Gamma}$, i.e. $s^2(1-\rho) \le \sqrt{m_{\text{eff}}/K}$, i.e. $\rho \ge \rho^\star_{\text{cov}}$—continuous in μ but non-analytic in ρ (the sharpness). For (ii), the single-planted-direction Le Cam second moment with a uniform prior on v is $\mathbb{E}_0[L_K^2] \to (1-\beta^2/\Gamma)^{-1/2}$, finite iff $\beta < \sqrt{\Gamma}$; a bounded second moment gives contiguity, hence no consistent test. Consistency checks: the μ=0 potential value $1+\sqrt{\Gamma}$ matches the Marchenko–Pastur edge, and the overlap matches BBP; both agree with the Stage-A simulation of §5. □

Scope of this appendix. This restates the wall and converse only—the two statements this paper most depends on—so the central claim is self-contained. The game value (Prop. 4), saturation (Prop. 5), and adaptive collapse (Prop. 6) transfer by the same fixed point under Lemma 1; their full proofs are in the core (or a future core), and their Stage-A confirmations are in §5. We present the CGMT reduction at the level of the standard Gordon substitution, not a self-contained regularity proof; its validity rests on the standard CGMT hypotheses plus the consistency checks and simulation above.

References (informal)

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